AWS Elastic Beanstalk

A Comprehensive Guide to Scalable Deployment Solutions

Mahendra Swain

1.Using Tomcat

Go to ElasticBeanStalk Console→

Step-1(Configure environment)

- Applications→Create Application→ Application name→Create environment→
- Web server environment→Environment name →Domain name → Platform(tomcat)→ Sample application→You can also Upload your code(Zip.file)→ Configuration:(Single instance (free tier eligible))→Next

Step-2(Configure service access)

(A).IAM Service Role :

Step-1

- AWS Service→TE=(Elastic Beanstalk)⍰ specified service(Elastic Beanstalk - Environment)→

Step-2

- role name→Attach permissions→(AWSElasticBeanstalkEnhancedHealth,→ AWSElasticBeanstalkService)→next→

Step-3

- Role name- aws-elasticbeanstalk-service-role →create role

- Attach Service Role

(B).EC2 instance profile Role:

Step-1

- Create Role → TE = EC2 →next

Step-2

- Permissions = AWSElasticBeanstalkMulticontainerDocker, AWSElasticBeanstalkWebTier,AWSElasticBeanstalkWorkerTier →Attach Policies

Step-3

- Role name- aws-elasticbeanstalk-ec2-role→Create Role

- EC2 instance profile Role →Select EC2 key pai →Next

Step-3(Set up networking, database, and tags)

- Optional→Next

Step 4(Configure instance traffic and scaling)

- Optional→Remove t3small →Next

Step-5(Configure updates, monitoring, and logging)

Optional

- System(Basic)→ Managed updates (Disable)→If required give email id →Next
-

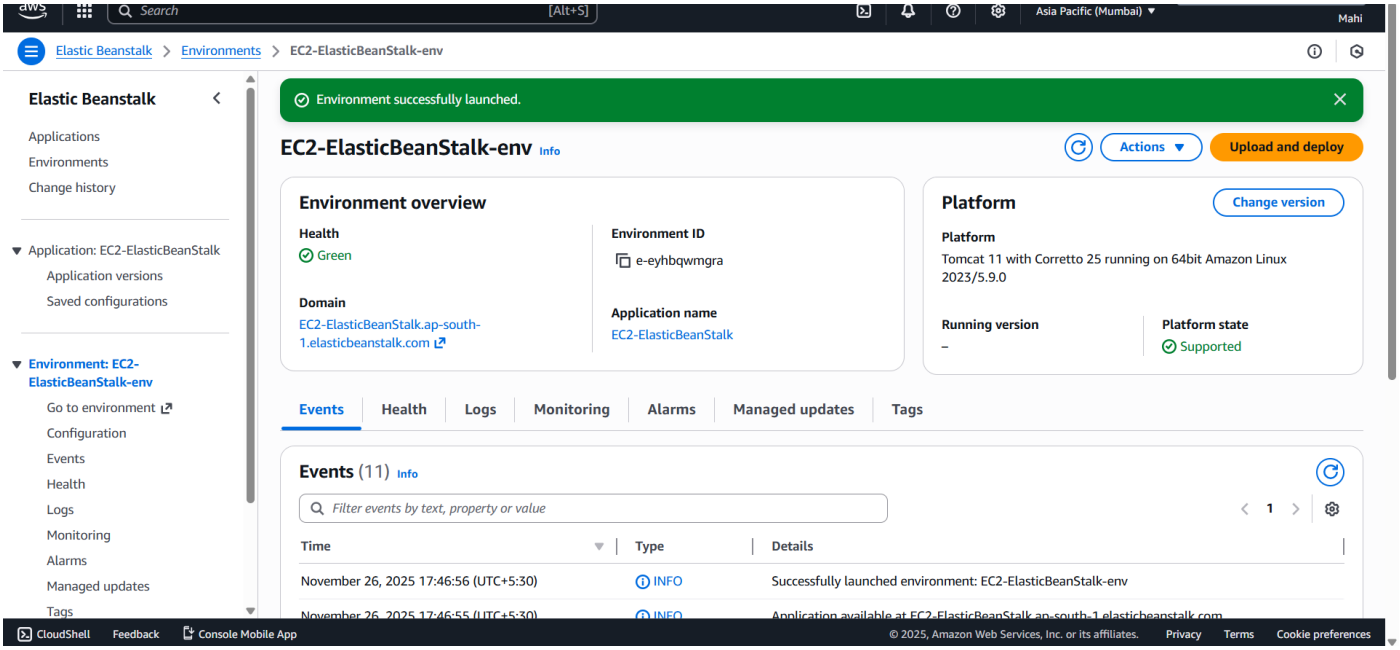
Step-6(Review)

- And Create

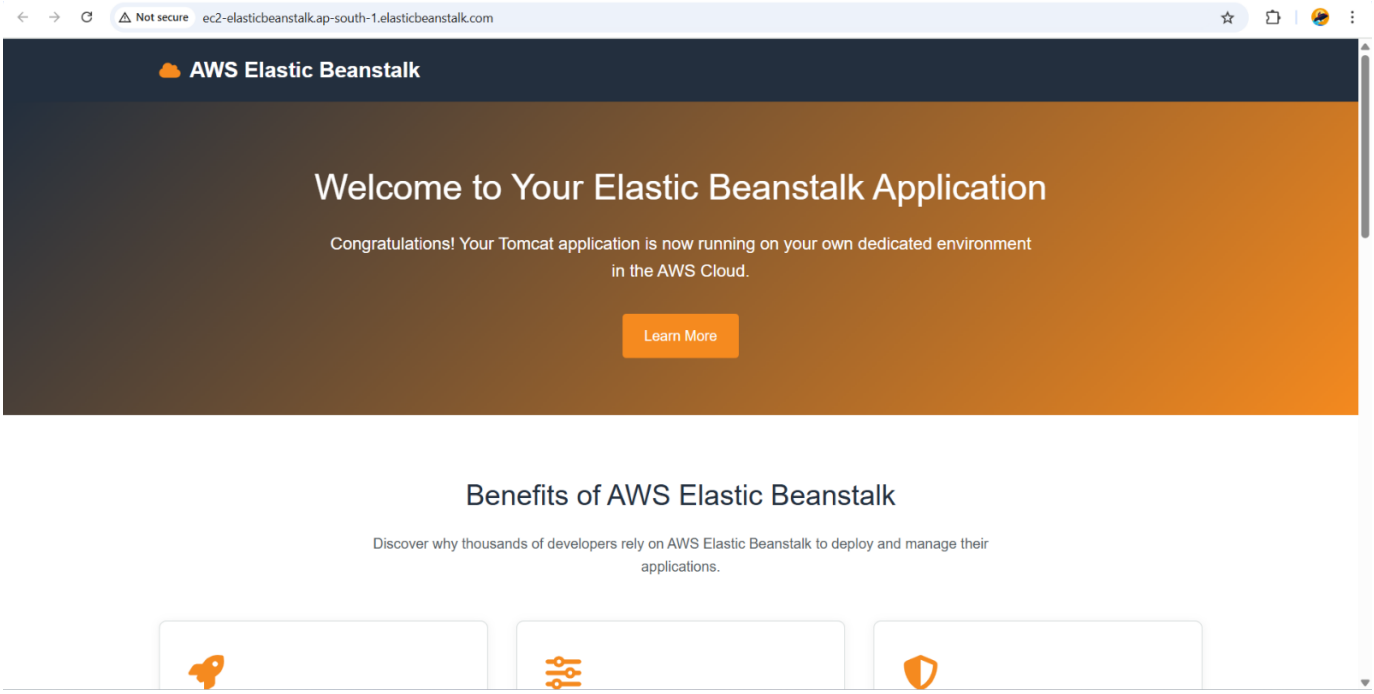
Environment overview:

- Health Status :Green

Now Environment Successfully Launched



Copy Domain URL:Open link in New Tab



2.Using Python

Go to ElasticBeanStalk Console→

Step-1(Configure environment)

- Applications→Create Application→ Application name→Create environment→
- Web server environment→Environment name →Domain name → Platform(**Python**)→ Sample application→You can also Upload your code(Zip.file)→ Configuration:(Single instance (free tier eligible))→Next

Step-2(Configure service access)

- Select Service role→EC2 instance profile Role →Select EC2 key pai →Next

Step-3(Set up networking, database, and tags)

- Optional→Next

Step 4(Configure instance traffic and scaling)

- Optional→Remove t3small →Next

Step-5(Configure updates, monitoring, and logging)

Optional

- System(Basic)→ Managed updates (Disable)→If required give email id →Next
-

Step-6(Review)

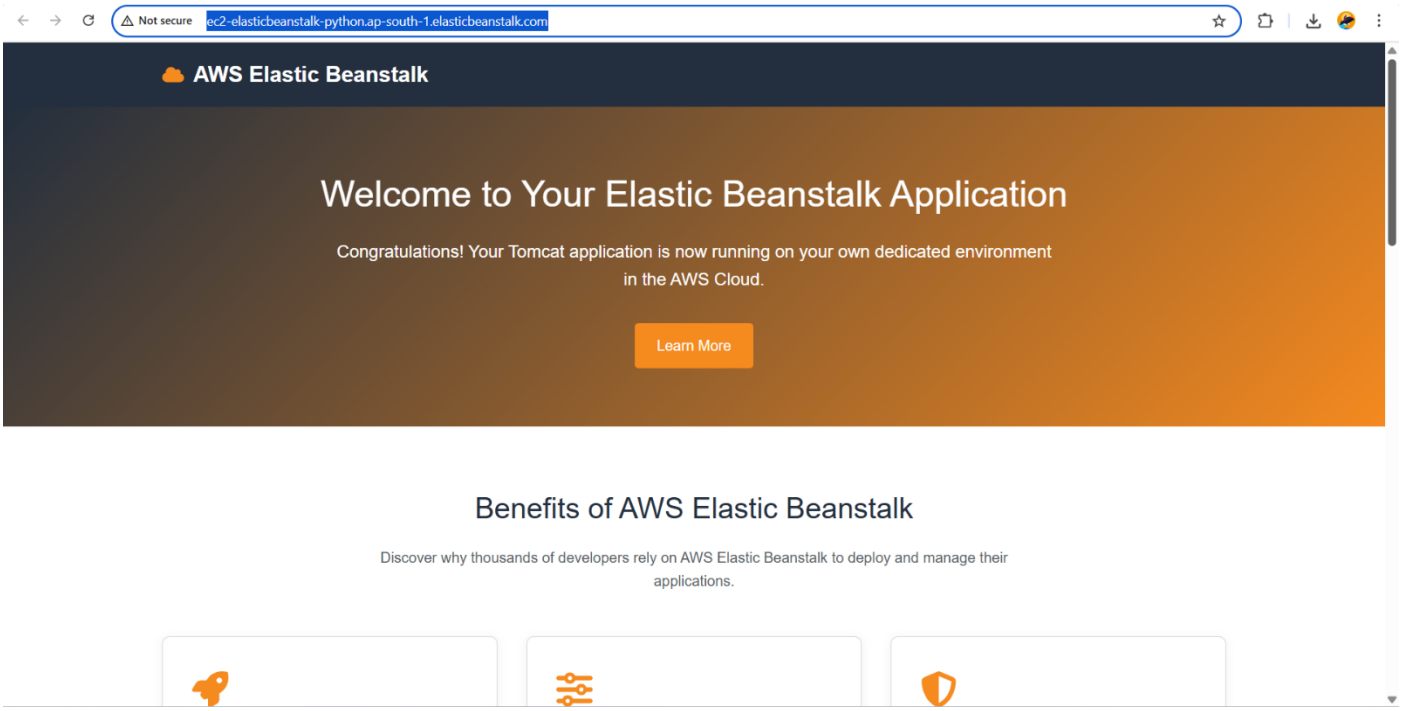
- And Create

SWAP

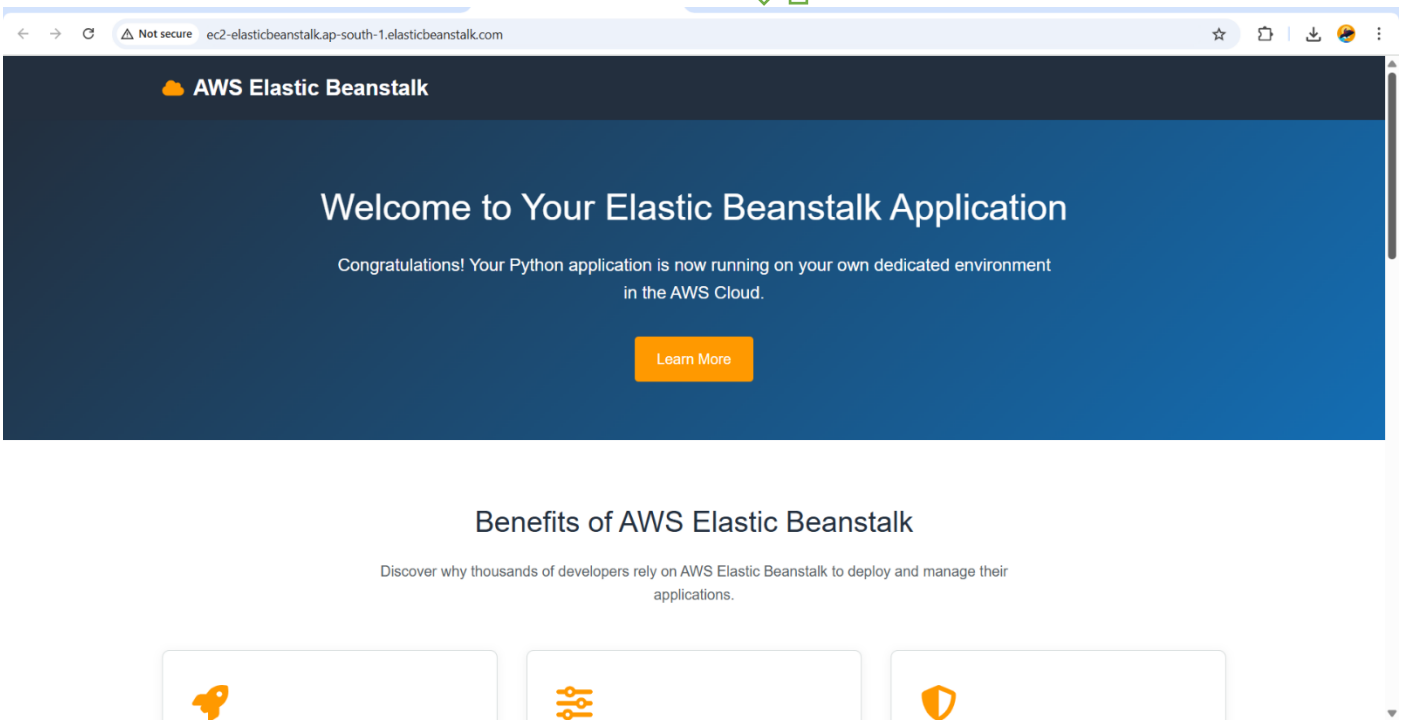
Go to ElasticbBeanStalk console →

- Go to Application→ Select Tomcat Environment →Go to Action→Select Swap Environment Domain
- Select python Environment →Swap

Swap Tomcat ⇄ **Python** Successfully



Now Without Any Downtime Successfully Deploy



Tomcat(HA)

Go to ElasticBeanStalk Console→

Step-1(Configure environment)

- Applications→Create Application→ Application name→Create environment→
- Web server environment→Environment name →Domain name → Platform(Tomcat)→ Upload your code(Zip.file)→ Configuration: **High availability**→Next

Step-2(Configure service access)

- Select Service role→EC2 instance profile Role →Select EC2 key pai →Next

Step-3(Set up networking, database, and tags)

- Optional→Next

Step 4(Configure instance traffic and scaling)

- Specify Availability Zones(1a and 1b)
- CPUUtilization
- Next

Step-5(Configure updates, monitoring, and logging)

Optional

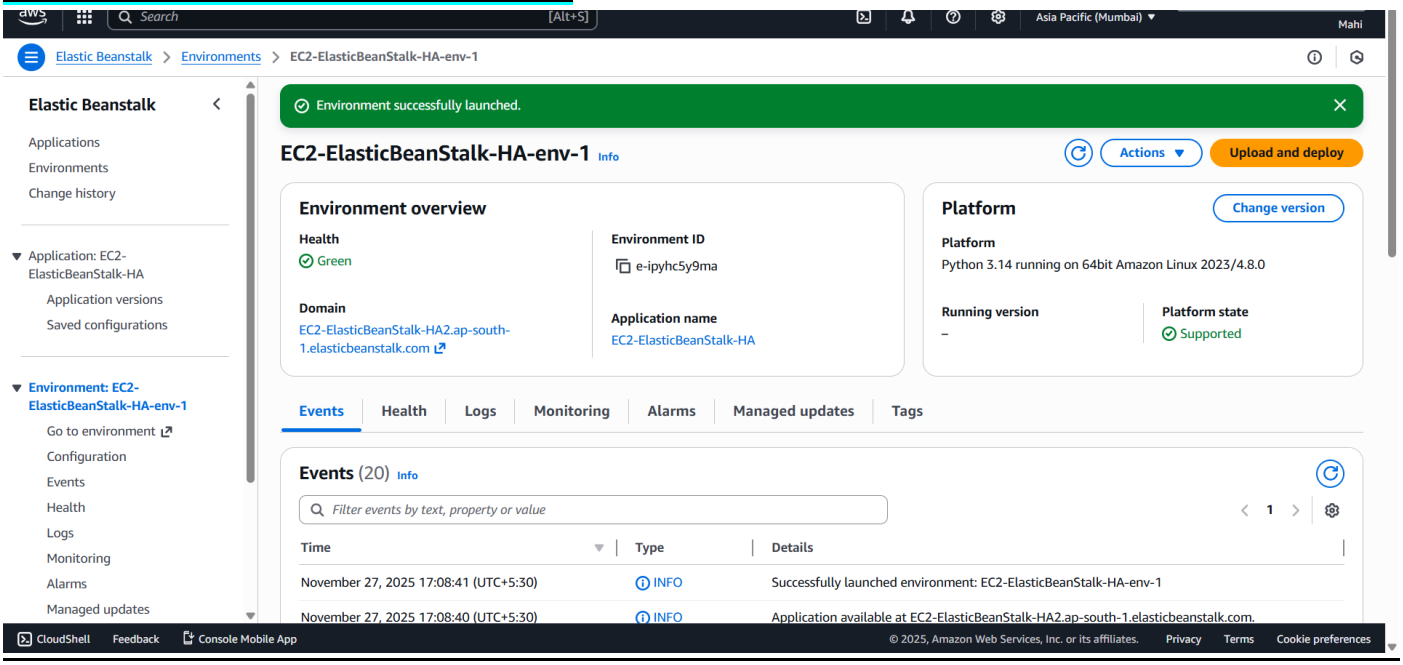
- System(Basic)→ Managed updates (Disable)→If required give email id →Next

Step-6(Review)

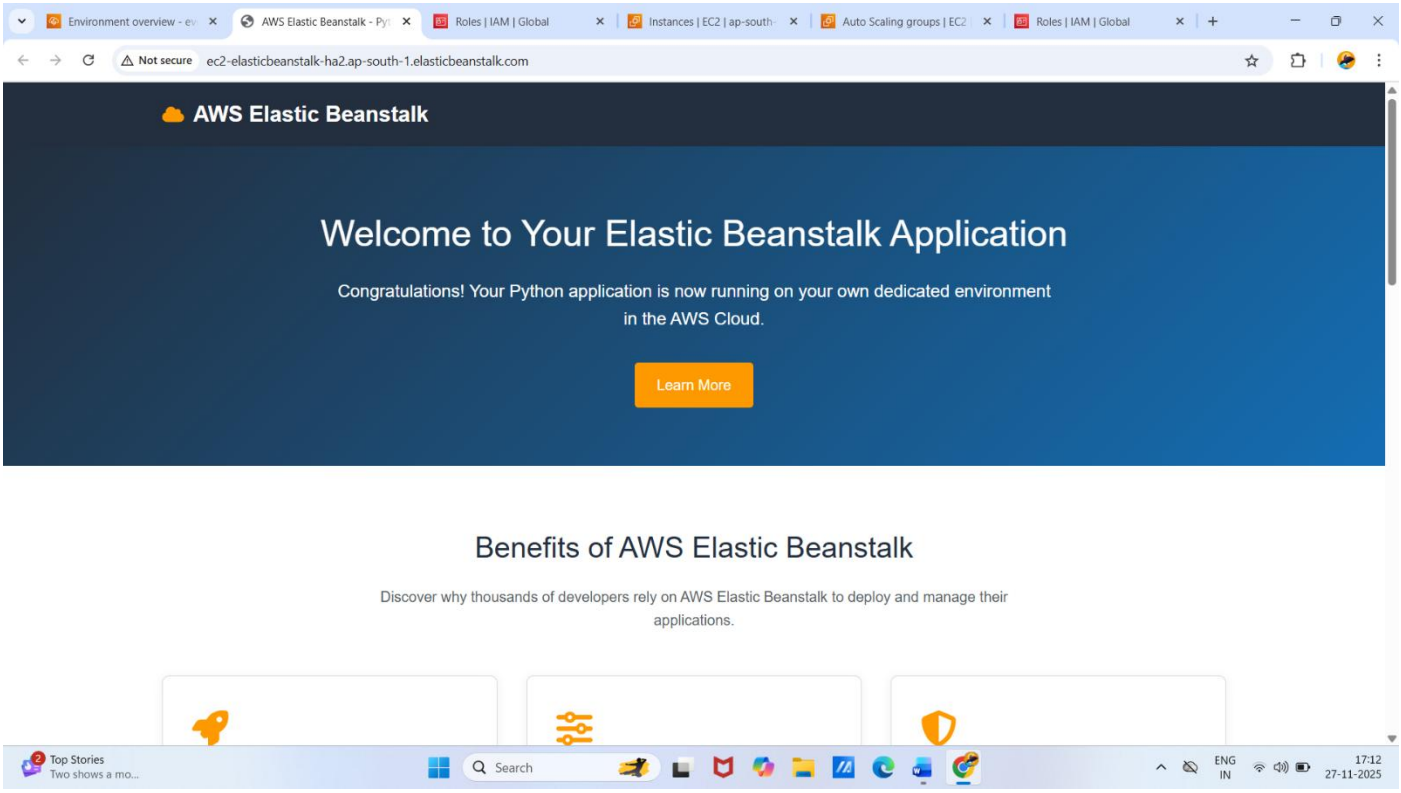
- And Create

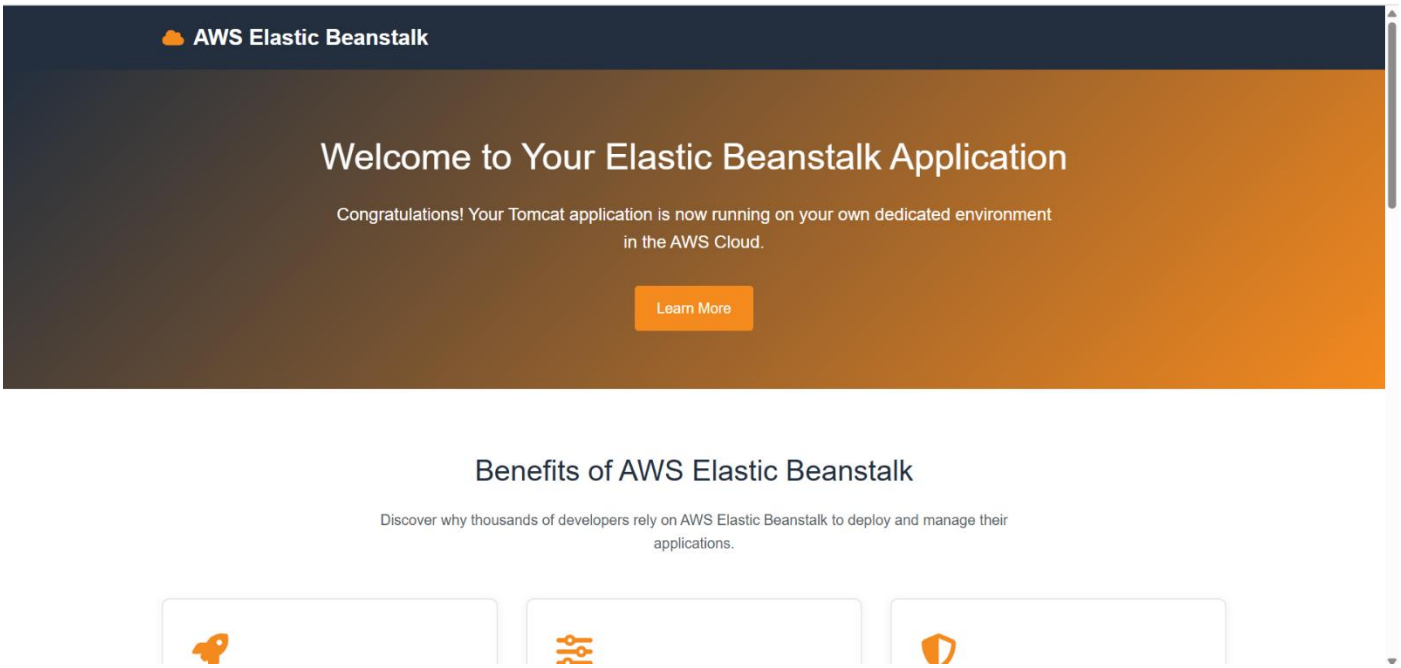
Python Environment with HA : With a Same Procedure

New Application and environment with HA



Blue & Green deployment On HA Preset



Tomcat ⇅ Python

Without any Downtime Successfully Deploy